SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Strength of Materials: Stress, Strain, SFD/BMD & Deflection of Beams

Department: CIVIL	**Course / Subject:** Strength of Materials (CE401PC)
Semester: Semester 4	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Hooke's Law, Poisson's ratio, Shear Force Diagrams (SFD), Bending Moment Diagrams (BMD), Flexural bending formula, Torsion of shafts, and Euler column buckling.

Unit 1: Simple Stresses and Strains

Tensile, compressive, shear stresses, True vs engineering stress-strain curve, Modulus of Elasticity, Poisson's ratio, Volumetric strain, Thermal stresses in bars.

Unit 2: Principal Stresses & Mohr's Circle

Stresses on inclined planes, Normal and tangential stresses, Maximum shear stress, Mohr's circle graphical construction, Theories of elastic failure.

Unit 3: Shear Force and Bending Moment

Types of beams (Cantilever, Simply supported, Overhanging), Point of contraflexure, Relation between load, shear force, and bending moment, SFD/BMD derivations.

Unit 4: Flexural & Shear Stresses in Beams

Pure bending theory, Flexure formula (M/I = sigma/y = E/R), Section modulus of rectangular and I-sections, Shear stress distribution across cross-sections.

Unit 5: Torsion of Shafts & Deflection of Beams

Torsion equation (T/J = tau/r = G theta/L), Power transmitted by shafts, Double integration method, Macaulay's method for beam deflection, Euler's column buckling formula.